

Test and Measurement Tools for LED Lighting



Abhishek A. Mutha
is a senior technical
correspondent at EFY

In the arena of light emitting diode (LED) test and measurement (T&M), manufacturers are creating many modular T&M devices, which allow for a swift assessment of lighting components and devices. Drivers are an important component in the performance and reliability of LEDs as well. There is a wide range of equipment available in the market for testing just the LED drivers. Let us take a look at some of the latest T&M systems available for LEDs and the lighting segment in general.

Modular, miniature spectrometer for lighting T&M

Miniature spectrometers are enabling new applications and implementations like quality control and product analysis in the lighting industry. These spectrometers provide flexible solutions for a wide range of upcoming applications in the LED industry today.

Developers and manufacturers of lighting solutions harness the power of a spectrometer to solve a variety of problems. Particularly, with respect to LEDs, engineers utilise it in applications that require test-retest reliability in industrial or other environments with varying conditions such as LED binning and process monitoring. Be it LEDs for exterior lighting or just daily quality control of LED production, a spectrometer should measure quickly and be reliable.

With a range of 190nm to 1100nm and a resolution of 0.1nm, Ocean Optics' flame spectrometer (Fig. 1) was announced earlier this year as a device that can be configured to specific application needs with user-interchangeable slits to adjust for varying specifications. These slits allow for variation in resolution and throughput of the spectrometer, as required. Small, rugged and easy-to-use, this spectrometer has a thermal stability of 0.05nm per centigrade up to 650nm range, which is an important feature for LED and other light measurement.

A trend that has been making waves in other test equipment like thermal imagers is the availability of miniature devices that latch on to a smartphone to function in a sort of symbiotic relationship. It seems this trend is catching on in this space too with devices like AsenseTek's Lighting Passport (Fig. 2). It is a professional-grade spectrometer that connects to any smartphone running either Android or iOS, while being small enough to fit in a pocket. There seems to be very little trade-off on functionality, too; firms using this device have worked on projects like Shanghai Formula 1 Racetrack and National Grand Theatre in Beijing.

One device that is often compared to Lighting Passport is UPiTek MK350, which is a somewhat older device, but this also makes it much more mature and tested, and therefore has a lot of positive reviews across the Web favouring it.

Similarly, another handheld spectrometer from GL Optic can be used for research and quality control in the ever-expanding LED lighting industry. This analytical and mobile instrument helps engineers measure a variety of parameters such as luminous flux (lumen), correlated colour temperature, illumination value (lux), colour coordinates and colour rendering



Fig. 1: Ocean Optics' flame spectrometer



Fig. 2: AsenseTek's Lighting Passport